AKTU NEXTGEN ENGINEERING

WHATSAPP GRP

1 & 2 YR

3 & 4 YR

FREE COURSES

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

ITCS

Page No.:
Date: / /

ONESHOT + 3 PYQ Solution

UNIT-4 [Science, Management & Indian Knowledge System]

TOPICS :-

→ * Astronomy in India → [2021-22], [2022,23]
* Chemistry in Ancient India → [12
* Mathematics in Ancient India [2022-23] 5th sem
→ * physics in Ancient India [2022-23]

* Agriculture in Ancient India
→ * Medicine in Ancient India [2022-23]
→ * Metallurgy in Ancient India [2022-23]
* Geography, Biology

* Harappan Technology
* Water Management in Ancient India [2021-22]
→ * Textile Technology in India [2022-23] [2021-22]
* Writing Technology in India.

→ * Pyro Technics in India [2022-23] [5th sem]
* Trade in Ancient India

* India's Dominance up to Pre-Colonial Times.

YouTube channel: Multi Atoms

Page No.
Date / /

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

[Nov 2021-22]

* Describe the concept of Astronomy in India.

→ Indian astronomy has a rich history from prehistoric times to the Present

→ The movement of planets came to be emphasized and closely observed.

→ Rig veda calendar → a year of 360 days divide into 12 parts & a five-year yug to align lunar and solar year

→ Yajur Veda Calendar → Considers a lunar year of 354 days and a solar year of 365 days, dividing the year into six two-month seasons.

→ The first astronomical objects, like calendar sticks from the Andamans (12,000 years ago), noted lunar phases by a wooden stick.

→ Rock Art in Kashmir: Depictions such as a double sun and concentric circles, dating (back 7000 years), represent a supernova & meteor showers.

→ Mohenjo-daro-Ring-stones: found b/w 2600-1900 BCE, these artifacts with drilled holes are interpreted as devices to track sunrise times throughout the year.

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

[AKTU- 2022-23]

→ Explore the contribution of Aryabhata in Astronomy.

→ Jyotish Vedanga text established systematic categories in astronomy, but the more basic problem was handled by Aryabhata.

→ His Aryabhattiya, text is Concise, containing 121 verses

→ It includes sections on astronomical definitions, methods for determining the true position of planets, descriptions of the sun & moon's movements & calculations of eclipses.

→ Aryabhata deviated from Vedic astronomy, providing it with a scientific outlook that guided later astronomers.

---

* Indian Chemistry

→ The Indus valley civilization was the earliest society, which had developed & elaborate urban.

→ The Early chemistry in India begins from here.

→ Glass making, Pottery, Jewellery making, dying of clothes and tanning of dithers etc. were the major chemical arts and crafts in the later periods

→ the alchemical knowledge as per the activity expanded

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Major chemical products that contributed to development of chemistry in India.

1. Glass → main factory at Kopia (UP)
2. Paper → main centre in medival India.
3. Soap → Used plant like soap nuts (Ritha, Sikakai)
4. Dying → Mentioned in Atharva Veda.
5. Perfumes → Mentioned in Brihatsamhita text.
6. Ink → recipes found in Rasaratna kara.
7. Alcoholic Liquors → Somarasa in Vedas.

AKTU - 2022-23 (5th Sem)

Q. write short note on : mathematics in Ancient India.

→ The town planning of Harappa shows that the people possessed a good knowledge of measurement & geometry.

→ Mathematics originated from Sulva sutras;

→ Apastamba introduced practical geometry in the 2nd Century BCE. (include acute angle, obtuse angle & right angle)

→ Key Contribution → Notation System
→ Decimal System
→ Use of Zero.

→ Aryabhatta discovered algebra & formulated the area of a triangle, → development of trigonometry.

→ Key works → Surya Siddhanta, Brihatsamhita → for further research & Progress.

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.:
Date : / /

[AKTU 2022-23]

India's Contribution to Physics

→ From the Vedic times Indians has classified the material world into

(earth) (fire) (air) (water) (ether)
prithvi agni maya apa akasha.

→ Concept of parmanu :- Ancient Indian philosphers believed that except for ether, all elements comprised minuscule particles of matter. The smallest indivisible particle was termed parmanu.

→ Parmanu suggest Indian phisdophers had an abstract undurstanding of atomic division.

→ Vaisesika (Indian phisosaphical system) introduced the concept of parmanu.

→ Total Nine type of substance →
- five elements
- Time (Kala)
- Direction or Space (di)
- The mind (manas)
- The spirit (atman)

→ These substances had 24 Quality → fluidity, viscosity, Gravity, elasticity.

→ Characteristics of Sound, heat, Light also discussed.

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.
Date / /

* Agriculture in India

→ Agriculture in India dates back to the Indus Valley Civilization and even earlier in some parts of Southern India.

→ India has one of the largest expanses of agricultural & land globally

→ Good monsoon & extensive network of rivers.

→ In Agriculture forming the backbone of Economy & Civilization.

→ Key Crops → wheat., rice, millets, oilseeds, cotton, Sugar cane & Indigo.

→ Crops like tobacco, chillies, potato, guava, custard apple, cashew and pineapple were introduced in the 16th & 17th centuries.

→ The system of land measurement & irrigation developed.

* Medicine in Ancient India [AKTU - 2022-23]

→ Diseases, cure & medicines were mentioned for the first ~~time~~ time in the Atharva Veda.

→ Fever, Cough, consumption, diarrhoea, dropsy, Sores, leprosy and seizure are the disease mentioned.

→ The oldest medical system is ayurveda → Science of good health & longevity of life → using herbal medicine aims at removing the cause of diverse by striking at the root. → which is highly developed.

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

→ Centre of medicine & Learning at Takshila & Varanasi.

→ In Charaksamhita & Sushrut Samhita used Plant & herbs for medicine.

→ In Sarangdhara Samhita - use of opium (Afeem) for medicine.

Metallurgy in Ancient India. [AKTU-2022-23]

→ It can be defined as the extraction, purification, glowing & application of metals.

→ for over 7000 years, India has had a high tradition of metallurgical skills.

→ The potteries, bronze, and copper artifacts found in Indus valley civilization.

→ By the 1st Century AD, mass production of metals like iron, copper, silver, gold and of alloys like brass & bronze were taking place.

→ Harappens artefacts → arrowheads, axes, chisels, sickles, blades, needles, hooks & vessels.

→ Recent excavations in central parts of the Ganges valley shown that iron was produced early in 1800 BC.

→ Zinc, steel also is used in ancient India.

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.
Date / /

* Geography

→ Constant interaction b/w humans & nature led to the study of geography.
→ people understand their own physical geography & the western countries, they were unaware of their exact position on Earth & distances to other countries
→ Indians contributed in shipbuildings.
→ Lothal, a site in Gujarat proving that trade ~~flourish~~ flourished in those day by sea.
→ puranas also have info. about tirtha & tirtha yatra.

* Biology

→ Ayurvedic Medicines belong to the plant kingdom.
→ Ayurvedic text also deal with plants classification.
→ many refrences to plant diseases and their ~~tre~~ treat-ment are also available in the vedic literature.
→ Vinaya, the famous Buddhist text describe the blight & mildew diseases.
→ Germination → The technical term used for seed is vija. Germination of seed is called ankuriodbhudha.
→ Acc. to Susruta → proper season
→ good soil requisite supply of water
→ then germination will good.

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.:
Date : / /

* Harappan Technology of Ancient India

→ In early human stages, 'technology' refers to any modification of the natural environment, including stone tools, clothing & construction.

Harappan Civilization:

→ Agriculture was a mainstay.

→ ceramic technology also developed = pots, fine fired bricks.

→ Pots :- used for carrying water, storing seeds & grain and cooking
- wheel-turned, glazed or painted, covered with red slip.

→ Bricks :- fired bricks with a 1 x 2 x 3 proportion for stronger walls.
- Extensive use of baked / mud bricks & stone.

→ Bead-making : Perfected techniques for polishing, coloring, glazing, drilling & bleaching
→ fav stone → Carnelian, agate & Jasper.

→ Other Crafts : ~~Bone~~ → Stone & ivory Carving, carpet making, inlaid wood work.
→ Bangles made from gold, bronze.
→ Weaving with wheel-spun thread for cotton & silk.

YouTube Channel: Multi Atoms

SUBSCRIBE MULTI ATOMS

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.
Date

AKTU - (2022-23, 2021-22)

* **Textile technology in India.**

→ Early References

- Vedas mention various garments and fabrics like wool & silk, along with weaving and looms.
- Buddhist literature provides insights into weaving skills.
- Ajanta paintings depict clothing & weaving techniques, including dying, from 2,000 years ago.

→ Trade & Export.

- the Roman Empire saw India as a major exporter of textiles, especially cotton & silk.
- India exported cotton to China & silk to Indonesia.

→ Textile Specialties:

- Muslin → A thin, loosely woven cotton fabric suitable for hot climates, introduced to Europe from Bengal.
- Calico → A plain-woven textile made from unbleached cotton, originally from (Calicut), Kerala.
- Chintz → A type of calico printed with floral and other colorful patterns.
- Other fabrics :- Included useful textiles from hemp, flax, linen and Jute.

→ Technology → weaving → Dev. of complex looms with regional variations.
→ Dyeing → mineral sources and vegetal.

→ Blue from indigo
- Red from plants like madder
- Yellow from turmeric, mango bark.
- Black from iron acetate.

YouTube Channel: Multi Atoms

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.:
Date:

* **Writing Technology in India.**

- Birch Bark : Used mainly in north India, with ink made from finely ground charcoal powder & gum or soot from oil lamp.
- Palm leaves : a sharp point was used to tear the leaf's surface film, use charcal powder.

→ In both the cases, considerable skills were developed to preserve manuscripts from insects and fungi.

→ The paper introduced in the 11th Century CE in India from China via Nepal.

→ By the late 15th Century, Kashmir produced high-quality paper from rags and hemp, with lime and soda added to whiten the pulp.

→ Several Europeans visitors from the 15th to 18th Century testify that Indian paper was of high quality.

[AKTU - 2021-22]

* **Water Management in Ancient India**

→ Ancient India showcased a variety of water structures for harvesting, conserving & distributing water efficiently.

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.
Date / /

→ Harappans used trapezoids bricks to construct wells, preventing them from collapsing underground pressures.

→ wells were built in various shapes (circular, square, vertical, horizontal) and materials (bricks, stone)

→ village ponds were crucial not just for ground water recharge but for being interconnected.

→ These network were maintained by village committees, ensuring equitable water distribution.

AKTU - 2022-23 [5th sem]

* Pyrotechnics in India

→ Pyrotechnics is the science of using materials capable of exothermic chemical reactions for the production of heat, light, gas, smoke and sound.

→ fireworks appeared in India during the 13th or 14th century.

→ By the 16th century, gunpowder became a key element in warfare.

→ Sanskrit text describes the preparation of gunpowder using sulphur and charcoal in various ratios.
Saltpetre,

→ Rockets began to be used in indian wars from the 16th century onward.

→ The British took some of these rockets to England for study, which advanced rocket technology in European warfare

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.:
Date: / /

Trade in ancient India

→ The trade networks expanded significantly before and during Indus civilization

→ During the 1st millennium BCE, large cities such as Taxila, Ujjain, Mathura, Patna, Rajgir, Varanasi, Bharuch and Madurai emerged as major trade centers.

→ These cities were generally situated on major trading routes., well structured states provided the necessary infrastructure and conductive env. for promoting trade.

→ Kautilya's Arthashastra → lists trade as one of the three major economic activities.

→ The Harappans exported timber, semiprecious stone beads, shell bangles, ivory items & pearls to Oman, Bahrain and mesopotamia.

India's Dominance up to Pre-colonial times

→ From the onset of the historical period, India enjoyed a favourable trade balance, accumulating significant stocks of gold.

→ During Mughal times, India maintained trade dominance (Abu'l Fazl records 40,000 traders engaged in trade on the Indus & its Punjab tributaries).

Notes By Multi Atoms
Subscribe "Multi Atoms" YouTube Channel for More Subjects

Page No.
Date / /

→ Economic historians note India's high trade surplus with Europe, West Asia, & Africa in the centuries preceding British rule.

→ Because of India's efficient production of low-cost products like cotton & spices, alongside well-organised trading communities.

→ India & China controlled nearly 60% of the world's GDP 2,000 years ago, remaining premier economic & trade powers until the colonial era.

→ Colonial rule coincided with a steep decline in India's overseas trade dominance & overall production as trade and industry rules were dictated by the colonial powers.

Subscribe
Multi Atoms
Join
Telegram for Notes

AKTU NEXTGEN ENGINEERING

WHATSAPP GRP

1 & 2 YR

3 & 4 YR

FREE COURSES